College of Engineering and Technology - Smart Village

Computer Engineering Department

ECE3502
Systems Programming

Lecture 2 –Language Representation

Regular expressions

Regular expressions provide a way to **define patterns over an alphabet Σ**, and the **languages** defined by these patterns are called **regular languages**. Using operations like **union**, **concatenation**, and **closure**, you can **construct complex languages** from simple building blocks

- **Set-theoretical union (+):** Represents a choice between elements.

  Example:

  Regular expression: a + b

  Language: {a, b} (*strings can be either a or b*).

- **Concatenation:** Joins elements in a sequence.

  Example:

  Regular expression: ab

  Language: {ab} (*only the string ab*).

Let's See Some Examples

- **Concatenation closure (*):** Represents repetition (*zero or more times*).

  Example:

  Regular expression: a*

  Language: {δ, a, aa, aaa, ...} (*any number of a's, including zero*).

Languages Definition

A Regular expression can describe a Language

Language → Regular Expression

Language can be constructed from a regular expression / phrase

Phrase → Language

How this can be achieved?

Let's see some examples

Examples on constructing regular expression for a given language

Language → Regular Expression

	L= { 0. 1. 10, 100, 1000, 10000, ... }
	L={1, 01, 10, 010, 0010, ...}
	L= {δ , 0, 1, 01}

Where 0* ={δ, 0, 00, 000, 0000, ...}

Examples on constructing regular expression for a given language represented by a phrase (*indirect*)

Phrase → Language → Regular Expression

Examples: Let Σ = {0, 1}

So far, we **construct the language** (**set of strings**) for the **given phrase**, then **construct the regular expression**.

1. Phrase: 0 followed by any number 1's

Language L = {0 , 01, 011 , 0111, ……..}

Regular expression: 01*

2. Phrase: All strings of 0's and 1's, beginning with a 0

Language L = {0 , 01, 011 , 0111, …….., 00 , 000, 0000, …….}

Regular expression: 0(0 + 1)*

3. Phrase: All strings of 0's and 1's, ending with a 1

Language L = {1, 01 , 001, 0001 , 00001, ……, 11, 111, 1111, ……...}

Regular expression: (0 + 1)*1

4. Phrase: All strings of 0's and 1's containing at least one 0

Language L = {10, 01 , 101, 0101 , 010100, ……, 00, 000, 0000, ……...}

Regular expression: (0 + 1)*0(0 + 1)*

Can we construct the regular expression directly from a phrase

Examples on constructing regular expression for a given language represented by a phrase (*direct*)

Phrase → Regular Expression

<u>Examples:</u> Write the regular expression corresponding to the given phrase

1. The set of all words that **contains a's and b's of any length**

$$(a + b)^*$$

2. The set of all words that **begins with the symbol a and contain a's and b's of any length**

$$a(a + b)^*$$

3. The set of all words that **begins with symbol a and end with symbol b and contain a's and b's of any length**

$$a(a + b)^*b = a\ (\textit{arbitrary string of a's and b's})\ b$$

More examples on constructing regular expression for a given language represented by a phrase (*direct*)

Phrase → Regular Expression

Examples: Write the regular expression corresponding to the given phrase

1) All strings of 0's and 1's containing **at least two 0's**

   (0 + 1)*0(0 + 1)*0(0 + 1)*

   (0 + 1)*01*01*

Can you write another expression?

2) All strings of 0's and 1's containing **an even number of 0's**

   (1 + 01*0)*

   1*(01*01*)*

   (1*01*0)* 1*

Can we describe this phrase with other regular expression?

Therefore, there is no unique regular expression to represent a phrase that describes a language

Real examples on constructing regular expression for a given language represented by a phrase (*direct*)

Phrase → Regular Expression

Write a regular expression that represent the identifier in C language.

Knowing that:

Digit → 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
Letter → a|b|c|d…….|z|A|B|C|D……|Z
Underscore → _

What are the requirements for defining an identifier in C language

Identifer = (letter + _)(letter + Digit + _)*

Real examples on constructing regular expression for a given language represented by a phrase (*direct*)

Phrase → Regular Expression

Write a regular expression that represents an email address.

e.g. m.hassan@gmail.com

Knowing that:

Letter = {a, b, c, ….., z, .}

Letter* @ Letter*.Letter Letter Letter

Write a regular expression that represents a phone number having the format (123)-234-4567

Knowing that:

Digit = {0, 1, 2, …., 9}

(Digit Digit Digit) – Digit Digit Digit – Digit Digit Digit Digit

Or,

Digit = {0, 1, 2, …., 9}

Area = Digit Digit Digit

Exch = Digit Digit Digit

Phone = Digit Digit Digit Digit

(Area) – Exch – Phone

Is there another solution?

Deriving language From Regular Expression

Regular Expression → Language

Language Simplification Rules

1. $L(\delta) = \{\delta\}$
2. $L(a) = \{a\}$
3. $L(a\ b) = L(a)\ L(b)$
4. $L(a + b) = L(a) + L(b)$
5. $L(a^*)=L(a)^*$

Regular Expression Simplification Rules

1. $a\ a^* = a^*\ a$
2. $a^*\ a^* = (a^*)^* = a^*$
3. $(a\ b)^*\ a = a\ (b\ a)^*$
4. $(a+b)^* = (a^*\ b^*)^* = (a^* + b^*)^* = a^*(ba^*)^*$

Examples on
Deriving language From Regular Expression

Example 1

Derive the language defined by the expression (a+b)a*

Solution

L ((a+b)a*) = L(a+b)L(a*) ={a, b} L(a*)

={a , b} {δ, a ,aa, aaa, aaaa, ……}

= {a, aa, aaa, aaaa, aaaaa, ……..b, ba, baa, baaa, baaaa, …….}

Example 2

Derive the language defined by the expression ab*a

Solution

L (ab*a) = L(a)L(b*) L(a) ={a} L(b*) {a} ={a} (L(b))* {a}

= {a} {δ, b, bb, bbb, bbbb, ……}{a}

= {a, ab, abb, abbb, abbbb, ……..}{a}

= {aa, aba, abba, abbba, abbbba, ...}

Examples on
Deriving language From Regular Expression

Example 3

Derive the language defined by the expression a*b*

Solution

L(a*b*) = L(a*) L(b*) = {δ, a, aa, aaa,……} L(b*)

={δ, a, aa, aaa,……} {δ, b, bb, bbb,……}

={δ, b, bb, bbb,…,a, ab, abb, abbb,…, aa, aab, aabb, aabbb,…, aaa, aaab, aaabb,…}

Example 4

Derive the language defined by the expression (ab)*

Solution

L((ab)*) = (L(ab))* = {δ, ab, abab, ababab, abababab,…}

As,

{δ, b, bb, bbb,…,a, ab, abb, abbb,…, aa, aab, aabb, aabbb,…, aaa, aaab, aaabb,…}

≠ {δ, ab, abab, ababab, abababab,…}

Therefore, L(a*b*) ≠ L((ab)*)

Examples on
Deriving language From Regular Expression

Example 5

Derive the language defined by the expression (a+b)*(a+bb)

Solution

L ((a+b)*(a+bb)) = L((a+b)*) L(a+bb) = (L(a+b))* L(a+bb)

=(L(a)+L(b))*L(a+bb)

={a, b}* L(a+bb)

= {δ, a, aa, aaa, aaaa,…, b, bb, bbb, bbbb, ….}L(a+bb)

= {δ, a, aa, aaa, aaaa,…., b, bb, bbb, bbbb,….}{a, bb}

= {a, bb, aa, abb, aaa, aabb, aaaa, aaabb,…,ba, bbb, bba, bbbb, bbba, bbbbb,…}

Example 6

Derive the language defined by the expression (aa)*(bb)* b

Solution

L ((aa)*(bb)* b) = L((aa)*) L((bb)*) L(b) = (L(aa))*(L(bb))*L(b)

= {δ, aa, aaaa, aaaaaa,….} (L(bb))*L(b)

= {δ, aa, aaaa, aaaaaa,…. } {δ, bb, bbbb, bbbbbb,…….} L(b)

= {δ, aa, aaaa, aaaaaa,…. } {δ, bb, bbbb, bbbbbb,…….} {b}

= {δ, bb, bbbb,…, aa, aabb, aabbbb,…, aaaa, aaaabb, aaaabbbb, ….}{b}

= {b, bbb, bbbbb,…, aab, aabbb, aabbbbb,…, aaaab, aaaabbb, aaaabbbbb,…}

Grammar

- A **meta-language** is a language used to **define other languages.**
- A **grammar** is a **meta-language** used to **define the syntax of a language**.

Therefore, **meta-language** provides the **tool to describe** the **grammar** of a language

Example:

Suppose that we have the following grammer

<expression> ::= <term> | <term> "+" <expression>
<term> ::= <number> | <number> "*" <term>
<number> ::= "0" | "1" | "2" | ... | "9"

- It consists of:
  - Finite set of **terminal symbols**
  - Finite set of **non-terminal symbols**
  - Finite set of **production rules**
  - **Start symbol**

Grammar

Example: Validate the following expression 3 + 2 * 5, given the following set of rules (**grammar**)

<expression> ::= <term> | <term> "+" <expression>
<term> ::= <number> | <number> "*" <term>
<number> ::= "0" | "1" | "2" | ... | "9"

Solution:

<expression>
→ <term> "+" <expression>
→ <number> "+" <expression>
→ 3 + <term>
→ 3 + <number> "*" <term>
→ 3 + 2 * <number>
→ 3 + 2 * 5

Can you Solve this problem?

Context-Free Grammar (CFG)

It is the grammar that consists of a **finite set of grammar rules** in the form of **quadruple** (**N**,**T**, **P**, **S**).

Where:

- **N** is a set of non-terminal symbols.
- **T** is a set of terminals where **N** $\cap$ **T** = **NULL**
- **P** is a set of rules, **P**: **N** $\rightarrow$ (**N** $\cup$ **T**)*
- **S** is the start symbol

Example:

1. The grammar ({A}, {a, b, c}, P, A), P: A $\rightarrow$ aA, A $\rightarrow$ abc
   e.g. abc, aabc, aaabc, aaaabc
2. The grammar ({S}, {a, b}, P, S), P: S $\rightarrow$ aSa, S $\rightarrow$ bSb, S $\rightarrow$ δ
   e.g. δ, bb, aaaa, bbbb, abba, aaaaaa, bbbbbb, abaaba
3. The grammar ({S, F}, {0, 1}, P, S), P: S $\rightarrow$ 00S | 11F, F $\rightarrow$ 00F | δ
   e.g. 000011, 110000, 00001100, 11 (*all strings must start with 00 or 11*)

Examples on
Deriving language ,Regular Expression, and CFG from a phrase

Example 1: Given the following phrase that represent a language L.

All possible strings over {a, b} that begin with an a and end with b

1) Find the regular expression that represent the above language
2) Construct the CFG that represent the above language

Solution:

1) Regular Expression

L = {ab, aab, abb, aaab, abbb, ……}

a(a + b)*b

2) Construct the CFG

$S \rightarrow aXb$

$X \rightarrow aX \mid Xb \mid \delta$

Example 2: Write a Grammar for unsigned decimal integers (*e.g. 10, 99, 34, 111, ….*)

Solution:

Integer $\rightarrow$ Digit Integer | Digit

Digit $\rightarrow$ 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

Examples on
Constructing CFG for a language represented by phrase

Example 1: Write a CFG for C-language identifiers which are strings of letter, digits, and underscore. The identifier must start with a letter.

Solution:

ID $\rightarrow$ Letter A
A $\rightarrow$ Letter A | _A | Digit A | δ
Letter $\rightarrow$ a | b | c | d ……. | z | A | B | C | D …… | Z
Digit $\rightarrow$ 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

Example 2: Given that $\Sigma = \{0, 1\}$, construct context free grammar for each of the following phrases:

1) All strings over Σ that starts and ends with the same symbol

Solution:

$S \rightarrow 0A0$
$S \rightarrow 1A1$
$A \rightarrow 0A \mid 1A \mid \delta$

2) All strings over Σ that has an odd length

Solution:

$S \rightarrow A$
$A \rightarrow 1A \mid 0A \mid \delta$

Is that right? NO

$S \rightarrow A$
$A \rightarrow 1B \mid 0B$
$B \rightarrow 1A \mid 0A \mid \delta$

3) All strings over Σ that has an odd length and its middle symbol is 0

Solution:

$S \rightarrow A0A$
$A \rightarrow 1A \mid 0A \mid \delta$

Examples on
Constructing CFG for a language represented by Regular Expression

Example 1: Find a CFG that generates the language.

$$L = \{a^n b^m c^m d^{2n} \mid n \geq 0, m > 0\}$$

For n=0 and m>0

Number of a's and d's is zero and the string contains only b's and c's having the same number

For n>0 and m>0

Number of d's is double the number of a's and the string containing a's, b's, c's, and d's having same number of b's and c's

Solution:

$S \rightarrow aAdd \mid bAc \mid bc$

Can we write as follows?

$S \rightarrow aAdd \mid bAc \mid \delta$

Violates the condition m>0

Example 2: Find a CFG that generates the language.

$$L = \{a^n b^m \mid 0 \leq n \leq m \leq 2n\}$$

For n=0

m>=n ⇒ m>=0, however, m<=2n ⇒ m<=0 ⇒ m=0

Therefore, number of a's and b's is zero (empty string)

For n>0

m>=n or m<=2n

Therefore, number of b's may be equal or greater than number of a's (at maximum=2n)

Solution:

$S \rightarrow aSb \mid aSbb \mid \delta$